
The Energy of Memorization: Why Graph-Level Label Noise Sharpens Representations and How to Stop It

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Graph Neural Networks (GNNs) are vulnerable to label noise in graph classification, yet the mechanisms driving this vulnerability are poorly understood. We
2 empirically identify a property of the *learned representations* that tracks noise
3 memorization: the within-graph Dirichlet energy of node features. Energy decreases as the model captures genuine class structure, then rises sharply, mainly in
4 high-frequency components, when the model begins to memorize corrupted graph
5 labels. We support this observation with a Poincaré–bi-Lipschitz lower bound on
6 the energy implied by noise fitting, combined with an entropy-based asymmetry
7 result adapted from Qian et al. [2025] indicating that the energy growth is preferentially absorbed by graphs most informative for classification. Guided by these
8 observations we evaluate three interventions: a Dirichlet-energy regularizer ($\mathcal{L}_{DE}$),
9 a robust loss (GCOD), and a spectral weight constraint (CE+W2). Across eight
10 benchmarks, including the full and modified ogbg-ppa, symmetric noise up to 60%,
11 asymmetric noise, and a transfer to Cora node classification, the three interventions
12 all reduce within-graph energy and improve robustness while preserving clean-data
13 performance. GCOD, our practical recommendation, matches OMG [Yin et al.,
14 2023] at $\sim 70\times$ lower training overhead.
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18 1 Introduction

19 Graph Neural Networks (GNNs) are now the workhorse of *graph classification*, the task of assigning
20 a single label to an entire graph, with applications spanning molecular property prediction [Wale and
21 Karypis, 2006], bioinformatics [Borgwardt et al., 2005], and social-network analysis [Yanardag and
22 Vishwanathan, 2015a]. In real deployments, the labels attached to graphs are routinely noisy: assays
23 disagree, annotators err, and weak supervision injects systematic mistakes. Although learning under
24 label noise has been thoroughly studied for images [Algan and Ulusoy, 2021] and for node classification [Dai et al., 2021, Yuan et al., 2023a], graph classification under noisy labels is dramatically less
25 explored [NT et al., 2019, Yin et al., 2023].
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27 **Graph classification is a fundamentally different setting.** In node classification, label noise
28 corrupts *individual node labels* on a single graph, and homophily, the tendency of connected nodes
29 to share labels, is a structural prior that directly couples Dirichlet energy of the *label signal* to
30 noise. In graph classification, label noise flips the label of an *entire graph*: nodes have no individual
31 classification labels, and within-graph homophily/heterophily is largely orthogonal to the graph-level
32 label. The standard tools and intuitions from node classification therefore do not transfer.

Our central observation. We measure the Dirichlet energy of *learned node representations*, not of labels. Empirically, this energy first decreases as the model captures genuine, transferable patterns; then, when the model begins to memorize incorrect graph-level labels, it rises sharply, with the surge concentrated in the high-frequency spectrum. This rise is not a tautology: noise is injected only at the graph-label level, while node features and edges are unchanged, so any correlation between energy and labels has to emerge through training dynamics. Theorem 1 supports this observation analytically: under a bi-Lipschitz regularity assumption, fitting noisy graph-level labels for two structurally similar graphs forces a lower bound on their within-graph Dirichlet energies via the Graph Poincaré inequality.

Why does suppressing this energy growth help? Combining a dataset-level spectral identity (Proposition 1) with an entropy argument from Qian et al. [2025], Section 6 argues that uncontrolled energy growth is preferentially absorbed by *high-entropy graphs*, the ones carrying most discriminative signal at pooling. Reducing within-graph Dirichlet energy therefore plausibly suppresses noise fitting while preserving the high-entropy regions classification accuracy depends on, a hypothesis we test with three mechanistically distinct interventions.

Contributions.

- A controlled study of *when* GNNs fit graph-level label noise (Section 4), isolating three failure modes: over-parameterization, low-order graphs, and small training sets.
- Within-graph Dirichlet energy as a diagnostic for graph-level noise memorization (Section 5), with a frequency decomposition that localizes the surge to high-frequency components and an empirical scaling with the noise rate.
- Analytical support for the diagnostic (Section 6): a spectral identity (Proposition 1), a Poincaré–bi-Lipschitz lower bound under noise fitting (Theorem 1), and an entropy-based asymmetry result (Corollary 1). These results are unique to graph classification, where labels and feature smoothness live on different objects, unlike node classification.
- Three interventions (Section 7): a Dirichlet-energy regularizer $\mathcal{L}_{DE}$, a centroid-based loss GCOD, and a spectral weight constraint CE+W2. They are mechanistically distinct and all reduce within-graph energy while improving robustness; GCOD is our practical recommendation.
- Evaluation on eight benchmarks under symmetric, asymmetric, and high-rate (60%) noise, the full and a modified ogbg-ppa, and a transfer to Cora node classification. GCOD matches OMG [Yin et al., 2023] and SOP [Liu et al., 2022] at $\sim 70\times$ lower compute.

2 Related Work

Learning under label noise. Robust loss functions [Ghosh et al., 2017, Patrini et al., 2017], sample re-weighting [Liu and Tao, 2015, Liu et al., 2022], early-learning regularizers [Arpit et al., 2017], and small-loss filtering [Gui et al., 2021] dominate the image-classification literature. Wani et al. [2023] introduce NCOD, a centroid-based outlier-discounting loss that we adapt to graphs.

Noisy-label learning on graphs. Most prior work concerns *node* classification, exploiting homophily, contrastive losses, or pseudo-labels [Dai et al., 2021, Yuan et al., 2023a,b, Li et al., 2024, Kang et al., 2018]. For *graph* classification, NT et al. [2019] propose a surrogate loss that discards estimated-noisy labels under specific assumptions, and OMG [Yin et al., 2023] combines contrastive learning, MixUp [Lim et al., 2022], and curriculum filtering. We differ in two ways: (i) we provide a spectral/energy-based explanation of *why* GNNs fit graph-level noise, rather than only mitigating it, and (ii) our practical method (GCOD) achieves better or comparable robustness at a fraction of the compute.

Dirichlet energy and over-smoothing. Dirichlet energy quantifies signal smoothness on graphs [Zhou and Schölkopf, 2005]. Existing analyses establish that GNNs act as low-pass filters whose energy decays exponentially with depth under spectral conditions [NT and Maehara, 2019, Cai and Wang, 2020, Oono and Suzuki, 2020, Rusch et al., 2023], and that weight-matrix eigenvalues control attraction/repulsion of node features [Giovanni et al., 2023]. Energy-based regularizers [Zhou et al., 2021, Bo et al., 2021, Chen et al., 2023] address *too little* energy (over-smoothing) on *node* classification. We address the opposite regime: *too much* within-graph energy caused by graph-level label noise. Qian et al. [2025] recently show that, for graph classification specifically, over-smoothing in *high-entropy* regions is the most damaging; we leverage this insight (Corollary 1) to argue why

suppressing energy growth is essential under graph-level noise. A more comprehensive related-work discussion is in Appendix A.

3 Background

Notation. A graph is $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$ with $N = |\mathcal{V}|$, degree matrix $\mathbf{D}$, adjacency $\mathbf{A} \in \{0, 1\}^{N \times N}$, feature matrix $\mathbf{X} \in \mathbb{R}^{N \times m}$, combinatorial Laplacian $\mathbf{L} = \mathbf{D} - \mathbf{A}$ (with eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N$) and normalized Laplacian $\Delta = \mathbf{D}^{-1/2} \mathbf{L} \mathbf{D}^{-1/2}$. We write $\lambda_2(\mathcal{G})$ for $\lambda_2(\mathbf{L})$ throughout and use Δ only when explicitly stated. In *graph classification*, the dataset is $\mathcal{D} = \{(\mathcal{G}^i, \mathbf{y}_i)\}_{i=1}^n$, $\mathbf{y}_i \in \{0, 1\}^{|C|}$ the one-hot graph label; we write c_i for the class. Under message passing, $\mathbf{H}_i^{l+1} = \text{UP}_{\Omega_i}(\mathbf{H}_i^l, \text{AGGR}_{\mathbf{W}_i}(\mathbf{H}_i^l, \mathbf{A}^i))$, with $\mathbf{H}_i^0 \equiv \mathbf{X}^i$ and $\mathbf{Z}^i \equiv \mathbf{H}_i^L$. A permutation-invariant readout $f_\theta : \mathbb{R}^{N_i \times m'} \rightarrow \mathbb{R}^{|C|}$ maps node features to class probabilities $\hat{\mathbf{y}}_i$. Under noisy labels, the observed c_i may differ from the ground truth; the test set is clean.

Dirichlet energy on graphs. For graph $\mathcal{G}^i$ with representation $\mathbf{Z}^i \in \mathbb{R}^{N_i \times m'}$ we use the combinatorial Dirichlet energy

$$E^{\text{dir}}(\mathbf{Z}^i) = \sum_{(u,v) \in \mathcal{E}^i} \|\mathbf{Z}_u^i - \mathbf{Z}_v^i\|_2^2 = \text{tr}((\mathbf{Z}^i)^\top \mathbf{L}^i \mathbf{Z}^i). \quad (1)$$

Small E^{dir} means smooth (low-frequency) representations within the graph; large E^{dir} means sharpening (high-frequency). The symmetrically-normalized variant $E_{\text{sym}}^{\text{dir}}(\mathbf{Z}) = \sum_{(u,v)} \|\mathbf{Z}_u / \sqrt{d_u} - \mathbf{Z}_v / \sqrt{d_v}\|_2^2 = \text{tr}(\mathbf{Z}^\top \Delta \mathbf{Z})$, used in some over-smoothing analyses [NT and Maehara, 2019, Oono and Suzuki, 2020], is equivalent up to degree-bounded constants and our qualitative claims hold for either; we use (1) because it pairs cleanly with the Poincaré inequality of Section 6. To track behavior across a class-restricted subset $\mathcal{S}$ of training graphs we use

$$E_{\text{set}}^{\text{dir}}(\mathcal{S}) = \frac{1}{|\mathcal{S}|} \sum_{\mathcal{G}^i \in \mathcal{S}} E^{\text{dir}}(\mathbf{Z}^i). \quad (2)$$

Crucially, (1) is computed on *learned representations*, not on labels. Within-graph homophily of features therefore evolves with training, decoupled from the graph-level label of $\mathcal{G}^i$.

4 When GNNs Fit Graph-Label Noise

GNNs trained with standard cross-entropy (CE) on graph classification tasks are not uniformly vulnerable to label noise. On the full ogbg-ppa [Szklarczyk et al., 2018a] (37 classes, the standard OGB benchmark), even 40% symmetric noise barely degrades CE accuracy, because the model is under-parameterized for the full task and behaves robustly [Zhang et al., 2021]. To expose the noise-fitting regime, we control three axes: **(F1)** number of classes (proxy for over-parameterization), **(F2)** graph order (size of each graph), and **(F3)** training-set size. We use a 5-layer GIN [Xu et al., 2019] with 300 hidden units throughout, unless stated.

In all three regimes (Fig. 1 and Appendix D), GNNs progressively memorize noisy graph-level labels. The natural next question is therefore: what changes in the network’s internal representations during memorization, and can this change be detected and controlled? The next section identifies Dirichlet energy as that internal signal.

5 Dirichlet Energy as a Diagnostic for Noise Fitting

We track $E_{\text{set}}^{\text{dir}}(\mathcal{D}_c)$, the within-graph energy averaged over training graphs of class c . Across architectures and datasets (Fig. 2), the diagnostic is consistent: $E_{\text{set}}^{\text{dir}}$ falls during the early “learning” phase, then rises precisely when the model begins to fit noisy graphs (CE-20% in Fig. 2b), with 40% noise yielding a strictly larger end-of-training energy than 10% noise (Appendix F).

Energy is sensitive to noise specifically through high-frequency components. A natural concern is that energy also rises during clean-data overfitting [Yan et al., 2022]; we confirm this in Appendix F but show the noise-driven surge is markedly steeper. To localize the surge spectrally, we use the

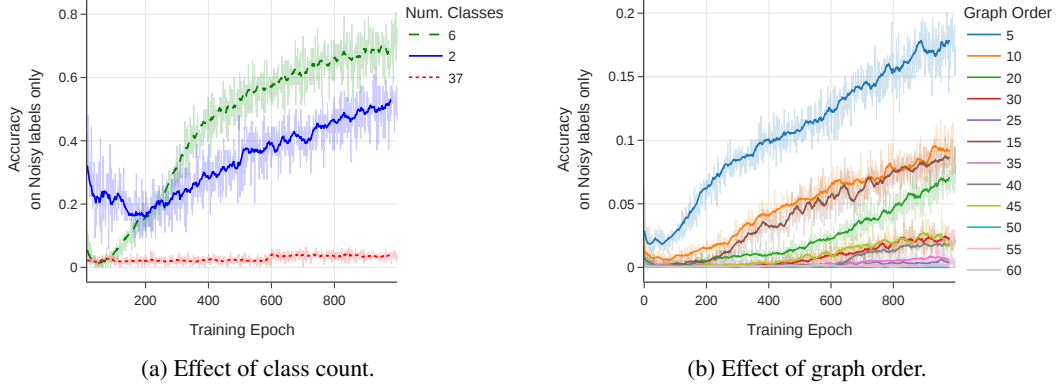


Figure 1: Training accuracy on *noisy labels only*: a higher curve means more memorization. (a) Reducing the number of PPA classes (with 20% symmetric noise) makes a fixed GIN over-parameterized, accelerating noise fitting. (b) Smaller average graph order (synthetic data, 35% noise) leaves less internal structure for averaging, enabling memorization.

127 HLFF-GNN decomposition [Xu et al., 2024] to split representations into low-frequency (Z_1) and
 128 high-frequency (Z_2) components. On ENZYMES with 30% noise, $E^{\text{dir}}(Z_1)$ remains essentially
 129 unchanged, while $E^{\text{dir}}(Z_2)$ rises monotonically, and slope/variance statistics of $E^{\text{dir}}(Z_2)$ provide an
 130 early warning for label noise (Appendix E). HLFF-GNN is used purely as an analytical tool; the same
 131 total-energy surge appears in standard GIN/GCN.

132 **Why energy is more useful than a generic overfitting signal.** Standard noise-detection signals
 133 require a *clean* held-out set: validation loss is uninformative when validation labels follow the same
 134 noise distribution as training (the noisy validation loss continues to decrease as the model fits noise,
 135 as we observe in our setting); train–test gap is unobservable at training time; small-loss heuristics
 136 need a calibration regime. Within-graph Dirichlet energy is computed entirely on the training graphs
 137 and their representations, so it remains a valid noise-fitting indicator under fully-noisy supervision.
 138 Beyond replacing the missing clean signal, it tells you *how* the network is failing: which spectral
 139 components drive the surge. This mechanistic specificity is what turns a diagnostic into a design
 140 principle, and motivates the three interventions in Section 7, each of which targets the same harmful
 141 spectral behavior in a distinct way.

142 6 Analytical Support: Why Energy Suppression Helps

143 Section 5 established Dirichlet energy as a reliable empirical signal. We now show this is not just
 144 correlation: under standard regularity, within-graph energy is *provably forced* to grow when a GNN
 145 fits graph-level label noise, with the growth absorbed disproportionately by structurally rich graphs.
 146 The argument combines three pieces: a dataset-level spectral identity (Proposition 1), the Graph
 147 Poincaré inequality (Lemma 1), and a bi-Lipschitz lower bound on the GNN (Lemma 2) [Chuang
 148 and Jegelka, 2022, Davidson and Dym, 2025, Juvina et al., 2024], yielding Theorem 1.

149 6.1 Three Building Blocks

150 **Proposition 1** (Dataset-level Dirichlet energy). *Let $\mathcal{D} = \{\mathcal{G}^1, \dots, \mathcal{G}^n\}$ with representations*
 151 *$\mathbf{Z}^i \in \mathbb{R}^{N^i \times m'}$ and adjacencies $\mathbf{A}^i$. Define the block-diagonal graph $\bar{\mathcal{G}} = (\bar{\mathbf{Z}}, \bar{\mathbf{A}})$ with $\bar{\mathbf{Z}} =$*
 152 *$[\mathbf{Z}^1 \parallel \dots \parallel \mathbf{Z}^n] \in \mathbb{R}^{N_{\text{tot}} \times m'}$ (vertical concatenation) and $\bar{\mathbf{A}} = \text{blkdiag}(\mathbf{A}^1, \dots, \mathbf{A}^n)$, with associated*
 153 *combinatorial Laplacian $\bar{\mathbf{L}} = \text{blkdiag}(\mathbf{L}^1, \dots, \mathbf{L}^n)$. Then*

$$E_{\text{set}}^{\text{dir}}(\mathcal{D}) = \frac{1}{|\mathcal{D}|} E^{\text{dir}}(\bar{\mathbf{Z}}) = \frac{1}{|\mathcal{D}|} \sum_{r=1}^{m'} \sum_{u=1}^{N_{\text{tot}}} \bar{\lambda}_u (\bar{\psi}_u^\top \bar{\mathbf{Z}}_{:,r})^2, \quad (3)$$

154 *where $\{(\bar{\lambda}_u, \bar{\psi}_u)\}$ are eigenpairs of $\bar{\mathbf{L}}$, and the spectrum decomposes as $\bar{\Lambda} = \bigcup_i \Lambda^i$ with each*
 155 *eigenvector supported on a single connected component $\mathcal{G}^i$.*

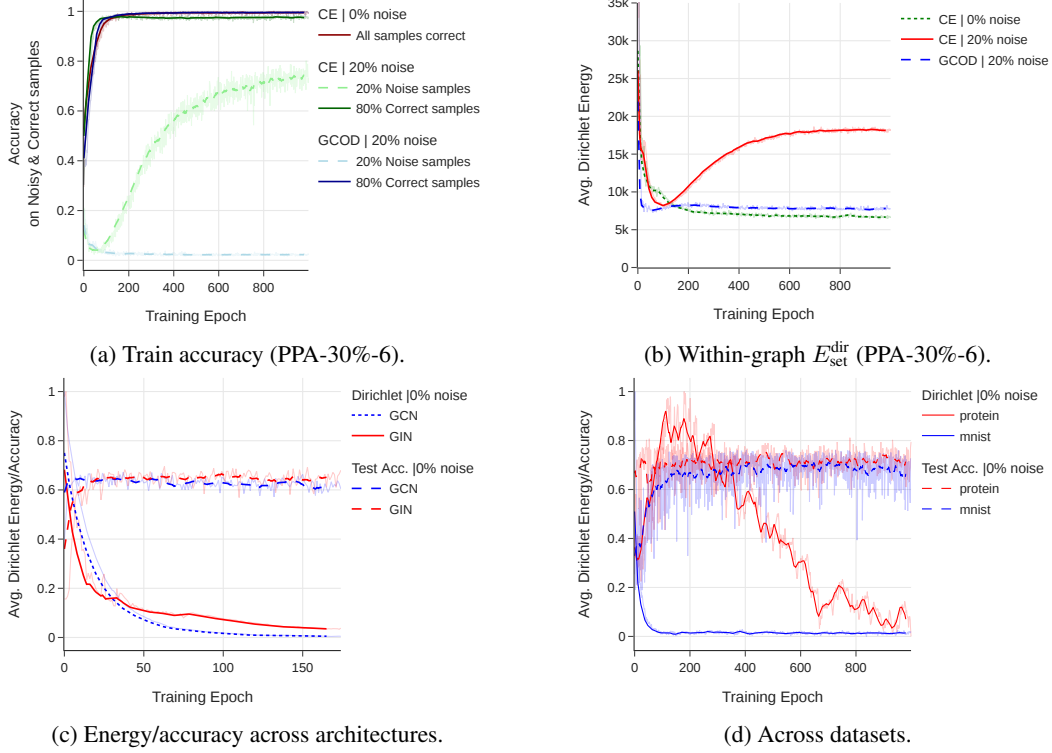


Figure 2: Within-graph Dirichlet energy of learned representations *rises during noise memorization*, across architectures (GIN/GCN) and datasets (PPA, MUTAG, MNIST, PROTEINS). Under clean labels, $E_{\text{set}}^{\text{dir}}$ steadily decays as accuracy improves; under noise, it follows the same decay until the model starts fitting noisy graphs, after which it sharply rises. GCOD suppresses this surge.

156 The proof (Appendix B) follows from the block-diagonal structure of $\bar{\mathbf{L}}$. The key structural fact: *each*
 157 *high-frequency eigenvector of $\bar{\mathbf{L}}$ is owned by exactly one graph*, so sharpening along it raises only
 158 that graph’s energy.

159 **Lemma 1** (Graph Poincaré, vector-valued form). *For any feature matrix $\mathbf{Z} \in \mathbb{R}^{N \times m'}$ on a connected*
 160 *graph $\mathcal{G}$ with combinatorial Laplacian $\mathbf{L} = \mathbf{D} - \mathbf{A}$ and second-smallest eigenvalue $\lambda_2(\mathcal{G}) > 0$, let*
 161 *$\bar{\mathbf{z}} = \frac{1}{N} \sum_v \mathbf{Z}_v$. Then*

$$\sum_{v=1}^N \|\mathbf{Z}_v - \bar{\mathbf{z}}\|_2^2 \leq \frac{1}{\lambda_2(\mathcal{G})} E^{\text{dir}}(\mathbf{Z}). \quad (4)$$

162 (4) bounds within-graph feature variance by Dirichlet energy scaled by the spectral gap; proof by
 163 Courant–Fischer per feature dimension (Appendix B).

164 **Lemma 2** (Bi-Lipschitz lower bound for a GIN block with 2-layer combine, adapted from Chuang and
 165 Jegelka, 2022, Davidson and Dym, 2025, Juvina et al., 2024). *Let $h_\theta(\mathbf{X}) = \sigma_2(\sigma_1(\mathbf{M}\mathbf{X}\mathbf{W}_1)\mathbf{W}_2)$*
 166 *be a single GIN aggregation $\mathbf{M} = (1 + \varepsilon)\mathbf{I} + \mathbf{A} \in \mathbb{R}^{N \times N}$ followed by a two-layer MLP with*
 167 *weights $\mathbf{W}_1 \in \mathbb{R}^{m \times m'}$ and $\mathbf{W}_2 \in \mathbb{R}^{m' \times m''}$ acting on the feature dimension, with the widening*
 168 *regime $m \leq m' \leq m''$ (so the linear maps have full column rank). Assume $\sigma_{\min}(\mathbf{M}) \geq \kappa > 0$ (e.g.,*
 169 *$1 + \varepsilon > |\lambda_{\min}(\mathbf{A})|$), $\sigma_{\min}(\mathbf{W}_k) \geq m_k > 0$, and that σ_1, σ_2 are leaky-ReLU with slope $\alpha > 0$ (or*
 170 *ReLU operated with a strict margin $\delta > 0$ along the line segment $\gamma(t) = \mathbf{X} + t\Delta\mathbf{X}$). Then*

$$\|h_\theta(\mathbf{X} + \Delta\mathbf{X}) - h_\theta(\mathbf{X})\|_F \geq c_{\text{lower}} \|\Delta\mathbf{X}\|_F, \quad c_{\text{lower}} = \alpha^2 m_1 m_2 \kappa. \quad (5)$$

171 A second aggregation step (turning the block into a true two-layer GIN $\sigma_2(\mathbf{M}\sigma_1(\mathbf{M}\mathbf{X}\mathbf{W}_1)\mathbf{W}_2)$)
 172 replaces c_{lower} by $\alpha^2 m_1 m_2 \kappa^2$; the qualitative content of the lemma is unchanged.

173 Proof: singular-value chain rule on the vectorized Jacobian $\mathbf{J} = \mathbf{D}_2(\mathbf{W}_2^\top \otimes \mathbf{I}_N) \mathbf{D}_1(\mathbf{W}_1^\top \otimes \mathbf{M})$
 174 (Appendix B). The lemma encodes that the GIN does not collapse meaningful feature differences, a
 175 property the model exploits when fitting noisy labels.

6.2 Main Theorem: Noise Fitting Forces Energy Growth

Consider two graphs $\mathcal{G}^i, \mathcal{G}^j$ with $N^i = N^j = N$ and similar pooled descriptors $\mathbf{g}_i \approx \mathbf{g}_j$ on the clean distribution (the same- N assumption is for notational simplicity; unequal N in Appendix B). Suppose label noise flips c_j to $c'_j \neq c_i$, forcing the network to output different graph-level labels. Let $\mathbf{h}_i \in \mathbb{R}^{N \times m'}$ be the post-message-passing signal of $\mathcal{G}^i$, $\mathbf{g}_i = \frac{1}{N} \sum_v \mathbf{h}_{i,v}$, and $\bar{\mathbf{h}}_i = \mathbf{1}_N \mathbf{g}_i^\top$.

Theorem 1 (Graph-level noise fitting forces E^{dir} to grow). *Under the hypotheses of Lemmas 1–2 and $N^i = N^j = N$,*

$$\|\mathbf{h}_i - \mathbf{h}_j\|_F \leq \sqrt{N} \|\mathbf{g}_i - \mathbf{g}_j\|_2 + \sqrt{\frac{E^{\text{dir}}(\mathbf{Z}^i)}{\lambda_2(\mathcal{G}^i)}} + \sqrt{\frac{E^{\text{dir}}(\mathbf{Z}^j)}{\lambda_2(\mathcal{G}^j)}}. \quad (6)$$

In particular, if the noisy training objective combined with an L -Lipschitz readout forces a graph-level output gap of at least Δ_{out} , so that $\|\mathbf{h}_i - \mathbf{h}_j\|_F \geq \Theta_{\text{noise}} := L^{-1} \Delta_{\text{out}}$, while the structural similarity gives $\|\mathbf{g}_i - \mathbf{g}_j\|_2 \leq \varepsilon$, then

$$\sqrt{\frac{E^{\text{dir}}(\mathbf{Z}^i)}{\lambda_2(\mathcal{G}^i)}} + \sqrt{\frac{E^{\text{dir}}(\mathbf{Z}^j)}{\lambda_2(\mathcal{G}^j)}} \geq \Theta_{\text{noise}} - \sqrt{N} \varepsilon. \quad (7)$$

Reading the bound. (7) lower-bounds the within-graph Dirichlet energies. The intuition is geometric: for two structurally similar graphs mapped to different labels, $\mathbf{g}_i \approx \mathbf{g}_j$, so the required output gap Θ_{noise} cannot come from the means alone; by Poincaré, the only remaining route is the within-graph variance E^{dir}/λ_2 . The unequal- N case (zero-padded matching) is handled in Appendix B. The bound has no node-classification analog: there labels live on the same nodes whose smoothness E^{dir} measures, so any link is trivial by construction.

Empirical anchor. The bound predicts that $\sqrt{E^{\text{dir}}/\lambda_2}$ should grow monotonically with the noise rate, since higher noise forces the encoder to discriminate more pairs of similar graphs. The PPA noise-rate sweep in Appendix F confirms this: asymptotic $E_{\text{set}}^{\text{dir}}$ at 10%, 20%, 30%, 40% noise is monotonically increasing.

6.3 Asymmetric Burden on High-Entropy Graphs

(7) only constrains the *sum* of $\sqrt{E^{\text{dir}}(\mathbf{Z}^i)/\lambda_2(\mathcal{G}^i)}$ across the two graphs. We next show that the structural-entropy ranking of Qian et al. [2025] dictates which graph absorbs the burden.

For a graph $\mathcal{G}$, the structural Shannon entropy is $H(\mathcal{G}) = -\sum_i \frac{d_i}{\sum_j d_j} \log \frac{d_i}{\sum_j d_j}$. Theorems 1–2 of Qian et al. [2025] prove that more nodes (at fixed sparsity) and denser connectivity (at fixed $|V|$) both strictly increase H .

Lemma 3 (Spectral counting). *The combinatorial Laplacian $\mathbf{L}^i$ has at most N^i distinct eigenvalues, contained in $[0, \max_{(u,v) \in \mathcal{E}^i} (d_u + d_v)]$, hence in $[0, 2d_{\max}(\mathcal{G}^i)]$ [Chung, 1997]. Equivalently, the normalized spectrum $\Lambda(\Delta^i) \subset [0, 2]$ with $\lambda_{\max}(\Delta^i) = 2$ iff $\mathcal{G}^i$ has a bipartite component. Therefore, when sharpening lifts spectral mass onto eigenvectors $\bar{\psi}_u$ of $\bar{\mathbf{L}}$ with $\bar{\lambda}_u \geq \tau$ for some threshold τ , the share allocated to $\mathcal{G}^i$ is bounded above by the count $n^i(\tau) = |\{j : \lambda_j^i \geq \tau\}|$, which is non-decreasing in N^i and in connection density (by Cauchy interlacing under vertex/edge addition).*

Corollary 1 (Asymmetric burden). *If $H(\mathcal{G}^j) \geq H(\mathcal{G}^i)$ via either of the conditions in Qian et al. [2025, Thm. 1–2], then the count of high-frequency eigenvalues of $\mathcal{G}^j$ above any threshold τ is at least that of $\mathcal{G}^i$, so the energy increase forced by Theorem 1 is preferentially absorbed by $\mathcal{G}^j$. This collapses the node-distribution distinctness of $\mathcal{G}^j$ [Qian et al., 2025, Sec. 3.3], the very property pooled graph classifiers rely on.*

Implication. Theorem 1 says noise fitting must raise energy somewhere; Corollary 1 says it raises it preferentially in the graphs the classifier most needs to distinguish. Suppressing within-graph energy growth therefore blocks the noise-fitting path while preserving the classification signal, an alignment that has no node-classification analog. This also explains why the surge of $E^{\text{dir}}(Z_2)$ under noise (Section 5) is steeper than under clean overfitting: noise fitting on similar graphs is structurally constrained to high-frequency, high-entropy directions, while clean overfitting can spread across the full spectrum.

7 Three Methods that Reduce Within-Graph Energy

The diagnostic and analytical results above suggest controlling within-graph Dirichlet energy through any of three mechanistically distinct levers: a regularizer on the energy itself, a loss that down-weights samples driving the energy up, or an architectural projection that removes the spectral directions responsible for sharpening. We instantiate one intervention per lever. GCOD is our practical recommendation; CE+W2 is included as a theoretical instantiation.

7.1 Explicit Dirichlet-Energy Regularization ($\mathcal{L}_{DE}$)

The most direct lever is to penalize $E^{\text{dir}}(\mathbf{Z}^i)$ above a class-aware threshold U_{c_i} :

$$\mathcal{L}_{DE}(\mathcal{D}) = \frac{1}{N} \sum_{i=1}^N [\max(0, E^{\text{dir}}(\mathbf{Z}^i) - U_{c_i})]^2, \quad \mathcal{L} = \mathcal{L}_{CE} + \lambda \mathcal{L}_{DE}. \quad (8)$$

We instantiate the threshold in two ways. **Class-specific:** U_c is recomputed every epoch as the mean E^{dir} over a small clean validation subset of class c . This adapts to class-dependent baseline energies (a molecular graph’s natural energy differs from a social graph’s), but requires clean held-out data. **Fixed:** $U_c = U$ globally, a single hyperparameter, no clean validation needed, at the cost of mild over-smoothing on clean labels. Both variants improve robustness; class-specific additionally improves *clean-label* accuracy by acting as a generalization-aware regularizer.

7.2 Graph Centroid Outlier Discounting (GCOD)

GCOD is our practical recommendation. It extends NCOD [Wani et al., 2023] (originally an image-classification loss) to graph classification, with two design changes that are necessary in the graph-classification regime: a third KL term that disentangles clean from noisy graphs via per-sample trainable parameters $\mathbf{u} \in [0, 1]^n$, and feedback through the current training accuracy a_{train} that prevents premature outlier discounting. For batch B with predictions $f_{\theta}(\mathbf{Z}_B)$, hard predictions $\hat{\mathbf{y}}_B$, soft labels $\tilde{\mathbf{y}}_B$ (centroid-based, see Appendix C.6), and observed labels $\mathbf{y}_B$:

$$\mathcal{L}_1 = \mathcal{L}_{CE}(f_{\theta}(\mathbf{Z}_B) + a_{\text{train}} \text{diag}_{\text{vec}}(\mathbf{u}_B) \mathbf{y}_B, \tilde{\mathbf{y}}_B), \quad (9)$$

$$\mathcal{L}_2 = \frac{1}{|C|} \|\hat{\mathbf{y}}_B + \text{diag}_{\text{vec}}(\mathbf{u}_B) \mathbf{y}_B - \mathbf{y}_B\|^2, \quad (10)$$

$$\mathcal{L}_3 = (1 - a_{\text{train}}) D_{KL}\{\mathcal{L}, \sigma(-\log \mathbf{u}_B)\}, \quad (11)$$

with $\mathcal{L} = \log \sigma(\text{diag}_{\text{mat}}(f_{\theta}(\mathbf{Z}_B) \mathbf{y}_B^{\top}))$. Updates are split: $\theta \leftarrow \theta - \alpha \nabla_{\theta}(\mathcal{L}_1 + \mathcal{L}_3)$, $\mathbf{u} \leftarrow \mathbf{u} - \beta \nabla_{\mathbf{u}} \mathcal{L}_2$. GCOD is not a theoretically derived energy minimizer. Empirically, however, it reduces $E_{\text{set}}^{\text{dir}}$ during training (Fig. 2) without explicitly penalizing it: by down-weighting graphs flagged as noisy via large u_i , it removes the gradient pressure that would otherwise push the network toward high-frequency directions, exactly as Theorem 1 predicts for any sample-level mechanism that suppresses noise-driven gradients.

7.3 Spectral Constraint on Weight Matrices (CE+W2)

A third lever, included as a theoretical instantiation, acts directly on the weights. Giovanni et al. [2023] show that for linear graph convolutions with symmetric $\mathbf{W}$, the gradient flow of E^{dir} decomposes along eigenvectors of $\mathbf{W}$: positive eigenvalues smooth, negative ones sharpen. Practical $\mathbf{W}_l^2$ is not symmetric, so we split $\mathbf{W}_l^2 = \mathbf{S}_l + \mathbf{N}_l$ and project only $\mathbf{S}_l$ to its positive eigenspace, leaving the skew $\mathbf{N}_l$ (purely imaginary spectrum, rotational) untouched. This is a heuristic extension of GRAFF — the energy of $\mathbf{MZW}$ depends on $\mathbf{W}^{\top} \mathbf{W}$, not just $\mathbf{S}_l$ — but the resulting CE+W2 needs neither clean validation nor extra hyperparameters and shows the energy mechanism can be controlled architecturally. Implementation details are in Appendix C.4; the per-epoch eigendecomposition is costly and can destabilize convergence, so we recommend $\mathcal{L}_{DE}$ or GCOD for practical use.

8 Experiments

Setup. We evaluate on ogbg-ppa [Szklarczyk et al., 2018a] (full and a controlled 30%/6-class subset), MUTAG, PROTEINS, ENZYMES, IMDB-BINARY, REDDIT-MULTI-12K, MSRC-21,

Table 1: Test accuracy on the PPA-30%-6 subset (5-layer GIN, mean $\pm$ std over 4 runs). **Best** is highlighted in red and bold, second-best in blue. “CE clean only” shows accuracy on the clean subset of the noisy data, as a reference upper bound. Class-specific $\mathcal{L}_{DE}^*$ requires a clean validation set.

Noise	Method	Test Acc.		Train Acc.	
		Best	Last	Best	Last
0%	CE	96.25 $\pm$ 0.05	91.25	100.0	99.33
	GCOD	96.65 $\pm$ 0.52	93.25	99.23	99.03
	CE+W2	96.50 $\pm$ 1.12	86.50	99.76	99.29
	$\mathcal{L}_{DE}$ (Fixed)	96.58 $\pm$ 0.27	85.70	99.26	98.29
	$\mathcal{L}_{DE}^*$ (Class-spec.)	96.96 $\pm$ 0.04	88.67	99.41	98.67
20%	CE (clean only)	96.15 $\pm$ 0.09	90.66	84.50	84.07
	CE	88.66 $\pm$ 0.16	62.58	94.98	93.45
	SOP	91.01 $\pm$ 0.44	85.50	79.17	77.64
	GCOD	93.91 $\pm$ 0.26	92.58	80.11	79.52
	CE+W2	89.83 $\pm$ 1.23	76.66	84.90	83.68
	$\mathcal{L}_{DE}$ (Fixed)	89.69 $\pm$ 0.57	80.25	78.07	70.14
	$\mathcal{L}_{DE}^*$	92.34 $\pm$ 0.41	80.92	84.55	83.73
40%	CE (clean only)	95.08 $\pm$ 0.13	80.44	68.15	67.05
	CE	82.08 $\pm$ 0.22	51.83	82.47	78.88
	SOP	82.33 $\pm$ 1.32	65.41	58.90	57.68
	GCOD	93.88 $\pm$ 0.04	91.08	65.09	64.31
	CE+W2	88.25 $\pm$ 1.13	56.33	68.78	65.88
	$\mathcal{L}_{DE}$ (Fixed)	88.58 $\pm$ 0.75	77.12	61.08	54.53
	$\mathcal{L}_{DE}^*$	88.55 $\pm$ 0.11	71.83	68.98	67.80

MNIST-graph, and Cora (node classification, transfer test). Backbones are GIN [Xu et al., 2019] and GCN [Kipf and Welling, 2017]; full hyperparameters are in Appendix C. We compare against three baselines: standard cross-entropy (CE); SOP [Liu et al., 2022], a strong noise-robust loss from image classification; and OMG [Yin et al., 2023], the leading graph-classification noise method. All numbers are mean $\pm$ std over 4 runs.

Why a controlled PPA subset? On the full 37-class PPA, all methods are under-parameterized for the task, and CE itself is robust (Section 4); a noise-robustness method cannot help where the baseline does not fail. We use a 30%/6-class PPA subset as a *controlled* testbed where over-parameterization triggers noise fitting, and we additionally report the full-PPA result to show our methods do no harm there. Generality claims are grounded in the seven additional benchmarks.

8.1 Main Result: Controlled PPA Subset

Table 1 reports the full ablation on the controlled PPA subset. GCOD is best at every noise level and, crucially, has the smallest gap between best and final test accuracy, indicating that it does not overfit late in training. CE+W2 and $\mathcal{L}_{DE}$ both improve robustness over CE, confirming the convergent-evidence prediction of Section 6. Under clean labels, $\mathcal{L}_{DE}^*$ even outperforms CE, a direct effect of the class-aware energy threshold.

8.2 Generalization Across Datasets and Noise Types

Table 2 reports GCOD across seven benchmarks beyond PPA. GCOD consistently improves over CE under both 20% symmetric and 40% asymmetric noise, and matches CE on REDDIT where the baseline is already saturated by class imbalance. The empirical trend that gains grow with the noise

Table 2: GCOD matches or beats CE across diverse datasets. Mean over 4 runs; $\pm$ std reported where available. *Sym*₂₀: 20% symmetric noise. *Asym*₄₀: 40% asymmetric noise. The *Asym*₄₀ GCOD column reports the std deviations from [Wani et al., 2023]’s rebuttal protocol.

Dataset	0% CE	Sym ₂₀ CE	Sym ₂₀ GCOD	Asym ₄₀ CE	Asym ₄₀ GCOD
PROTEINS	81.16	76.23	79.38	72.90	76.19 $\pm$ 1.22
MNIST	72.69	66.30	71.26	71.15	72.61 $\pm$ 0.43
ENZYMES	73.33	64.16	68.69	65.80	69.81 $\pm$ 0.98
IMDB-B	76.50	75.00	75.40	68.18	72.89 $\pm$ 0.91
MUTAG	94.73	89.47	90.01	91.16	93.19 $\pm$ 0.38
REDDIT	48.15	45.05	44.98	48.01	47.94 $\pm$ 2.87
MSRC-21	96.69	90.26	94.69	94.39	95.57 $\pm$ 0.32

Table 3: OMG protocol (Yin et al., 2023). Percentage gain over CE.

Dataset	OMG	GCOD
MUTAG	0.061	0.062
IMDB-B	0.047	0.049
PROTEINS	0.039	0.041

Table 4: Relative training time (CE-GIN = 1.00).

Method	Runtime
CE-GIN	1.00
SOP	1.048
GCOD	1.029
CE+W2	1.33
OMG [Yin et al., 2023]	$\approx$ 3.00

rate (Appendix F) is consistent with Theorem 1: the forced energy lower bound $\Theta_{\text{noise}} - \sqrt{N}\varepsilon$ scales with the noise level, so the room a method has to help also scales.

Full ogbg-ppa. Under the full 37-class benchmark, all methods including CE behave robustly to noise (Section 4); we therefore use it only as a do-no-harm check. Detailed results are in Appendix H.

Comparison with OMG. Under the OMG [Yin et al., 2023] protocol (Table 3), GCOD matches OMG’s accuracy (within 0.1–0.2 pp) while requiring $\sim 70\times$ less training overhead (Table 4): OMG adds $\approx 200\%$ runtime, GCOD only $\approx 2.9\%$.

Cross-setting transfer to node classification (Cora). Applied unchanged to node classification on Cora under 20% symmetric noise (full table in Appendix I, against 13 node-specific baselines including NRGNN, RTGNN, GNN-Cleaner, ERASE, CR-GNN), our methods rank in the top half on both GAT and GCN backbones: CE+W2 reaches 0.8157 (rank #2) on GAT and 0.7247 (rank #5) on GCN; GCOD reaches 0.7910 (rank #4) on GAT and 0.7403 (rank #4) on GCN. An energy-based regularizer that uses neither neighbor labels nor homophily yet outperforms specialized node-classification methods is direct evidence that the perspective captures a general property of GNN training under label noise.

Convergent evidence. The three methods are mechanistically distinct (architectural, regularization, sample re-weighting), yet all reduce $E_{\text{set}}^{\text{dir}}$ during training (Fig. 2, Appendix F) and improve robustness over CE — three independent levers acting on the same target, as Theorem 1 and Corollary 1 predict.

9 Conclusion

Within-graph Dirichlet energy of learned representations is an empirical signal of graph-level noise memorization, supported by a Poincaré–bi-Lipschitz lower bound and an entropy-based asymmetry argument specific to graph classification. The diagnostic translates into three interventions ($\mathcal{L}_{DE}$, GCOD, CE+W2) that all reduce within-graph energy and improve robustness across eight benchmarks while preserving clean-label performance; GCOD adds $\approx 3\%$ training overhead and matches OMG at $\sim 70\times$ lower compute. **Limitations.** The bound is a sufficient condition, not a closed-form characterization; we use synthetic noise protocols since real-world noisy graph-classification benchmarks are scarce; CE+W2 is a heuristic extension of GRAFF. **Broader impact.** Robust graph learning matters in chemistry and biology, where annotation errors are systematic, and GCOD avoids the need for clean validation data.

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A Additional Related Work

A.1 Learning under Label Noise

Beyond robust losses, additional families include sample relabelling [Arazo et al., 2019, Reed et al., 2015], two-network co-teaching and divide-and-mix approaches [Han et al., 2018, Li et al., 2020, Kim et al., 2023], and regularization through MixUp [Zhang et al., 2017]. Reweighting methods adapt sample importance via training dynamics or held-out probes [Liu and Tao, 2015, Pleiss et al., 2020]. Most of these were developed for image classification; their direct transfer to graph classification is non-obvious because graphs lack the strong feature regularities of images.

A.2 Graph Learning under Noise

Most graph-related noise work targets node classification, exploiting either homophily (RS-GNN [Dai et al., 2022], NRGNN [Dai et al., 2021], RTGNN [Yuan et al., 2023a], pairwise interactions [Du et al., 2023]) or self-supervision (pseudo-label denoising [Yuan et al., 2023b, Qian et al., 2023]). Edge/feature noise has also been studied, e.g. structural noise [Fox and Rajamanickam, 2019, Dai et al., 2022]. These methods typically assume connected nodes share labels, which is incompatible with our graph-classification setting where labels are graph-level. The seminal work of NT et al. [2019] treats graph classification under label noise via a surrogate loss; OMG [Yin et al., 2023] combines contrastive learning, MixUp, and curriculum filtering. Both are practical improvements but provide no spectral or energy-based explanation of *why* GNNs fit graph-level noise.

A.3 Dirichlet Energy and Spectral Properties of GNNs

NT and Maehara [2019] and Rusch et al. [2023] analyze GNNs as low-pass filters, with self-loops further shrinking eigenvalues. Cai and Wang [2020], Oono and Suzuki [2020] prove that GNN Dirichlet energy decays exponentially when the product of the largest singular value of the weight matrix and the largest Laplacian eigenvalue is below 1. Giovanni et al. [2023] disentangle smoothing/sharpening via positive/negative weight-eigenvalues. Zhou et al. [2021] (NeurIPS) uses energy regularization to prevent over-smoothing on *node* classification. Our setting is the opposite regime: too *much* energy under graph-level label noise.

Qian et al. [2025] introduce a structural-entropy viewpoint on over-smoothing for graph classification, showing that high-entropy regions are most damaged by over-smoothing. We invert and combine this with Proposition 1 to get Corollary 1: the graphs that suffer most from over-*sharpening* (the noise-fitting regime) are also the high-entropy graphs that carry the most discriminative signal. The two papers therefore identify symmetric ends of the same spectral failure.

A.4 Lipschitz/Stability Analyses

Chuang and Jegelka [2022] bound GIN outputs via the Tree Mover’s Distance, and Davidson and Dym [2025] extend uniform-Lipschitz analyses to MPNNs through Hölder continuity of aggregation/combination/readout. Juvina et al. [2024] derive optimal Lipschitz constants $\vartheta = \phi(\lambda_K)$ for graph convolutions under non-negative weights and ReLU. Gama et al. [2019] show output stability under graph-shift-operator perturbations. These results characterize stability w.r.t. inputs; we instead study stability against *label* noise, mediated by the within-graph energy of learned representations.

B Proofs and Extended Theory

B.1 Proof of Proposition 1

Let $\Lambda^i = \{\lambda_u^i\}_{u=1}^{N^i}$ be the spectrum of the per-graph combinatorial Laplacian $\mathbf{L}^i = \mathbf{D}^i - \mathbf{A}^i$. Define the block-diagonal Laplacian

$$\bar{\mathbf{L}} = \text{blkdiag}(\mathbf{L}^1, \dots, \mathbf{L}^n), \quad \det(\bar{\mathbf{L}} - \lambda \mathbf{I}_{N_{\text{tot}}}) = \prod_{i=1}^n \det(\mathbf{L}^i - \lambda \mathbf{I}_{N^i}), \quad (12)$$

so $\bar{\Lambda} = \bigcup_i \Lambda^i$. Let ψ_j^i be the j -th eigenvector of $\mathbf{L}^i$ at eigenvalue λ_j^i . The vector

$$\bar{\psi}_u = (0, \dots, 0, \psi_j^i, 0, \dots, 0)^\top \in \mathbb{R}^{N_{\text{tot}}} \quad (13)$$

(zero on every block other than i) satisfies $\bar{\mathbf{L}}\bar{\psi}_u = \lambda_j^i \bar{\psi}_u$. With $\bar{\mathbf{Z}} = [\mathbf{Z}^1 \parallel \dots \parallel \mathbf{Z}^n]$ (vertical concatenation), the combinatorial Dirichlet energy of the block-diagonal graph satisfies

$$E^{\text{dir}}(\bar{\mathbf{Z}}) = \text{tr}(\bar{\mathbf{Z}}^\top \bar{\mathbf{L}} \bar{\mathbf{Z}}) = \sum_{r=1}^{m'} \sum_{u=1}^{N_{\text{tot}}} \bar{\lambda}_u (\bar{\psi}_u^\top \bar{\mathbf{Z}}_{:,r})^2 = \sum_i \sum_{r,j} \lambda_j^i (\psi_j^{i\top} \mathbf{Z}_{:,r}^i)^2 = \sum_i E^{\text{dir}}(\mathbf{Z}^i), \quad (14)$$

where the second equality uses the spectral expansion in the orthonormal basis $\{\bar{\psi}_u\}$ and the third uses the block-supported structure of each $\bar{\psi}_u$. Hence $E_{\text{set}}^{\text{dir}}(\mathcal{D}) = \frac{1}{|\mathcal{D}|} \sum_i E^{\text{dir}}(\mathbf{Z}^i) = \frac{1}{|\mathcal{D}|} E^{\text{dir}}(\bar{\mathbf{Z}})$. $\square$

B.2 Proof of Lemma 1 (Graph Poincaré Inequality)

The combinatorial Laplacian $\mathbf{L} = \mathbf{D} - \mathbf{A}$ is real symmetric and positive semi-definite. Because $\mathcal{G}$ is connected, $\ker(\mathbf{L}) = \text{span}(\mathbf{1})$, and the eigenvalues are $0 = \lambda_1 < \lambda_2 \leq \dots \leq \lambda_N$. By the Courant–Fischer min–max characterization,

$$\lambda_2(\mathcal{G}) = \min_{\substack{\mathbf{x} \perp \mathbf{1} \\ \mathbf{x} \neq \mathbf{0}}} \frac{\mathbf{x}^\top \mathbf{L} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}}. \quad (15)$$

Fix any feature dimension r and let $\mathbf{z} = \mathbf{Z}_{:,r} \in \mathbb{R}^N$ with $\bar{z} = \frac{1}{N} \sum_v z_v$. The centered vector $\mathbf{z} - \bar{z}\mathbf{1}$ is orthogonal to $\mathbf{1}$, so

$$\lambda_2(\mathcal{G}) \|\mathbf{z} - \bar{z}\mathbf{1}\|_2^2 \leq (\mathbf{z} - \bar{z}\mathbf{1})^\top \mathbf{L} (\mathbf{z} - \bar{z}\mathbf{1}) = \mathbf{z}^\top \mathbf{L} \mathbf{z} = \sum_{(u,v) \in \mathcal{E}} (z_u - z_v)^2, \quad (16)$$

using $\mathbf{L}\mathbf{1} = \mathbf{0}$. The right-hand side is the per-feature-dimension contribution to $E^{\text{dir}}(\mathbf{Z})$ in (1). Summing over $r = 1, \dots, m'$ yields

$$\sum_{v=1}^N \|\mathbf{Z}_v - \bar{\mathbf{z}}\|_2^2 = \sum_{r=1}^{m'} \|\mathbf{z}^{(r)} - \bar{z}^{(r)}\mathbf{1}\|_2^2 \leq \frac{1}{\lambda_2(\mathcal{G})} \sum_{r=1}^{m'} \mathbf{z}^{(r)\top} \mathbf{L} \mathbf{z}^{(r)} = \frac{E^{\text{dir}}(\mathbf{Z})}{\lambda_2(\mathcal{G})}, \quad (17)$$

which is (4). The constant $1/\lambda_2$ is sharp, attained by the Fiedler vector ψ_2 . The corresponding bound for the normalized Dirichlet energy $E_{\text{sym}}^{\text{dir}}(\mathbf{Z}) = \text{tr}(\mathbf{Z}^\top \Delta \mathbf{Z})$ is obtained by the change of variable $\mathbf{y}_v = \mathbf{Z}_v / \sqrt{d_v}$ and applying the same argument with $\lambda_2(\mathbf{L})$ in the bound. $\square$

B.3 Proof of Lemma 2 (Bi-Lipschitz Lower Bound for a GIN)

The block under analysis is one GIN aggregation followed by a two-layer MLP combine,

$$h_\theta(\mathbf{X}) = \sigma_2(\sigma_1(\mathbf{M}\mathbf{X}\mathbf{W}_1) \mathbf{W}_2), \quad (18)$$

acting on $\mathbf{X} \in \mathbb{R}^{N \times m}$ with $\mathbf{M} \in \mathbb{R}^{N \times N}$ mixing rows, and $\mathbf{W}_1 \in \mathbb{R}^{m \times m'}$, $\mathbf{W}_2 \in \mathbb{R}^{m' \times m''}$ acting on the feature (column) dimension. For leaky-ReLU activations (or ReLU with a fixed margin $\delta > 0$ along $\gamma(t) = \mathbf{X} + t\Delta\mathbf{X}$, ensuring no piecewise-linear region change), each σ_k has derivative $\geq \alpha$ everywhere along the path.

Vectorized Jacobian. Using $\text{vec}(\mathbf{A}\mathbf{Y}\mathbf{B}) = (\mathbf{B}^\top \otimes \mathbf{A})\text{vec}(\mathbf{Y})$,

$$\begin{aligned} \text{vec}(\mathbf{M}\mathbf{X}\mathbf{W}_1) &= (\mathbf{W}_1^\top \otimes \mathbf{M}) \text{vec}(\mathbf{X}), \\ \text{vec}(\sigma_1(\mathbf{M}\mathbf{X}\mathbf{W}_1) \mathbf{W}_2) &= (\mathbf{W}_2^\top \otimes \mathbf{I}_N) \mathbf{D}_1 (\mathbf{W}_1^\top \otimes \mathbf{M}) \text{vec}(\mathbf{X}), \end{aligned}$$

so the full Jacobian in vectorized form is

$$\mathbf{J}(\mathbf{X}) = \mathbf{D}_2 (\mathbf{W}_2^\top \otimes \mathbf{I}_N) \mathbf{D}_1 (\mathbf{W}_1^\top \otimes \mathbf{M}), \quad (19)$$

with $\mathbf{D}_k = \text{diag}(\sigma'_k) \succeq \alpha \mathbf{I}$.

575 **Singular-value chain.** For any product $\mathbf{ABCD}$ where each factor has $\sigma_{\min} > 0$, $\sigma_{\min}(\mathbf{ABCD}) \geq$
576 $\sigma_{\min}(\mathbf{A})\sigma_{\min}(\mathbf{B})\sigma_{\min}(\mathbf{C})\sigma_{\min}(\mathbf{D})$ (chain rule for singular values; equivalent to applying the lower
577 bound $\|\mathbf{Ax}\| \geq \sigma_{\min}(\mathbf{A})\|\mathbf{x}\|$ inductively). Using $\sigma_{\min}(\mathbf{P} \otimes \mathbf{Q}) = \sigma_{\min}(\mathbf{P})\sigma_{\min}(\mathbf{Q})$,

$$\begin{aligned}\sigma_{\min}(\mathbf{J}) &\geq \sigma_{\min}(\mathbf{D}_2) \sigma_{\min}(\mathbf{W}_2^\top \otimes \mathbf{I}_N) \sigma_{\min}(\mathbf{D}_1) \sigma_{\min}(\mathbf{W}_1^\top \otimes \mathbf{M}) \\ &\geq \alpha \cdot m_2 \cdot \alpha \cdot m_1 \kappa = \alpha^2 m_1 m_2 \kappa = c_{\text{lower}}.\end{aligned}$$

578 **Fundamental theorem of calculus.** For $\gamma(t) = \mathbf{X} + t \Delta \mathbf{X}$, the strict-margin assumption ensures
579 that no entry of the pre-activation $\mathbf{W}_1 \mathbf{M} \gamma(t)$ or $\mathbf{W}_2 \sigma_1(\mathbf{W}_1 \mathbf{M} \gamma(t))$ changes sign along $[0, 1]$, so $\mathbf{D}_1$
580 and $\mathbf{D}_2$ are *constant* in t . Hence $\mathbf{J}(\gamma(t)) \equiv \mathbf{J}$ is constant, and $h_\theta \circ \gamma$ is exactly affine on $[0, 1]$:

$$\text{vec}(h_\theta(\mathbf{X} + \Delta \mathbf{X}) - h_\theta(\mathbf{X})) = \mathbf{J} \text{vec}(\Delta \mathbf{X}). \quad (20)$$

581 Taking the ℓ_2 norm (= Frobenius norm in matrix form) and applying $\|\mathbf{J} \mathbf{x}\|_2 \geq \sigma_{\min}(\mathbf{J}) \|\mathbf{x}\|_2$,

$$\|h_\theta(\mathbf{X} + \Delta \mathbf{X}) - h_\theta(\mathbf{X})\|_F \geq c_{\text{lower}} \|\Delta \mathbf{X}\|_F. \quad (21)$$

582 For a true two-layer GIN $\sigma_2(\mathbf{M} \sigma_1(\mathbf{M} \mathbf{X} \mathbf{W}_1) \mathbf{W}_2)$, the second Kronecker factor in the chain becomes
583 $(\mathbf{W}_2^\top \otimes \mathbf{M})$ instead of $(\mathbf{W}_2^\top \otimes \mathbf{I}_N)$, contributing an extra $\sigma_{\min}(\mathbf{M}) \geq \kappa$ and giving $c_{\text{lower}} =$
584 $\alpha^2 m_1 m_2 \kappa^2$. $\square$

585 **Remark 1** (Beyond strict margin). Without the strict-margin condition γ may cross piecewise-linear
586 boundaries of σ_k . Then $h_\theta(\mathbf{X} + \Delta \mathbf{X}) - h_\theta(\mathbf{X}) = (\sum_k p_k \mathbf{J}_k) \Delta \mathbf{X}$ with $\mathbf{J}_k$ the per-region Jacobians
587 (each satisfying $\sigma_{\min}(\mathbf{J}_k) \geq c_{\text{lower}}$) and $p_k \geq 0, \sum p_k = 1$. Convex combinations of matrices
588 with $\sigma_{\min} \geq c$ do not in general satisfy the same lower bound, so c_{lower} may degrade across region
589 boundaries. We therefore use Lemma 2 only as a local non-degeneracy regularity statement in
590 Theorem 1; this is sufficient for our purposes.

591 B.4 Proof of Theorem 1 (Noise Forces Energy Growth)

592 Let $\mathbf{h}_i, \mathbf{h}_j \in \mathbb{R}^{N \times m'}$ be the post-message-passing per-node signals of $\mathcal{G}^i$ and $\mathcal{G}^j$ (we assume $N^i =$
593 $N^j = N$; the unequal- N case is treated in the Remark in the main text). Let $\mathbf{g}_i = \frac{1}{N} \sum_v \mathbf{h}_{i,v} \in \mathbb{R}^{m'}$
594 be the mean-pooled graph descriptor, and let $\bar{\mathbf{h}}_i = \mathbf{1}_N \mathbf{g}_i^\top$ be its replication to a constant per-node
595 signal. By the Frobenius-norm triangle inequality,

$$\|\mathbf{h}_i - \mathbf{h}_j\|_F \leq \|\mathbf{h}_i - \bar{\mathbf{h}}_i\|_F + \|\bar{\mathbf{h}}_i - \bar{\mathbf{h}}_j\|_F + \|\bar{\mathbf{h}}_j - \mathbf{h}_j\|_F. \quad (22)$$

596 **Middle term.** Because both $\bar{\mathbf{h}}_i, \bar{\mathbf{h}}_j$ are constant signals of size $N \times m'$,

$$\|\bar{\mathbf{h}}_i - \bar{\mathbf{h}}_j\|_F^2 = \sum_{v=1}^N \|\mathbf{g}_i - \mathbf{g}_j\|_2^2 = N \|\mathbf{g}_i - \mathbf{g}_j\|_2^2, \quad (23)$$

597 so $\|\bar{\mathbf{h}}_i - \bar{\mathbf{h}}_j\|_F = \sqrt{N} \|\mathbf{g}_i - \mathbf{g}_j\|_2$.

598 **Variance terms.** Identifying $\mathbf{h}_i$ with $\mathbf{Z}^i$ (the layer- L representation whose Dirichlet energy we
599 measure), Lemma 1 applied to $\mathbf{Z}^i$ gives

$$\|\mathbf{h}_i - \bar{\mathbf{h}}_i\|_F^2 = \sum_v \|\mathbf{h}_{i,v} - \mathbf{g}_i\|_2^2 \leq \frac{E^{\text{dir}}(\mathbf{Z}^i)}{\lambda_2(\mathcal{G}^i)}, \quad (24)$$

600 and analogously for $\mathcal{G}^j$. Taking square roots and combining,

$$\|\mathbf{h}_i - \mathbf{h}_j\|_F \leq \sqrt{N} \|\mathbf{g}_i - \mathbf{g}_j\|_2 + \sqrt{\frac{E^{\text{dir}}(\mathbf{Z}^i)}{\lambda_2(\mathcal{G}^i)}} + \sqrt{\frac{E^{\text{dir}}(\mathbf{Z}^j)}{\lambda_2(\mathcal{G}^j)}}, \quad (25)$$

601 which is (6).

602 **Lower bound from the noisy training objective.** The graph-level prediction is $\hat{\mathbf{y}} = R(\mathbf{h})$ for some
603 permutation-invariant readout R that is L -Lipschitz in the Frobenius norm (mean- or sum-pooling
604 followed by an MLP with bounded weights). At convergence, the noisy training objective enforces
605 $\hat{\mathbf{y}}_i$ to lie near a one-hot vector at c_i and $\hat{\mathbf{y}}_j$ near a one-hot at $c_j \neq c_i$, so $\|\hat{\mathbf{y}}_i - \hat{\mathbf{y}}_j\|_2 \geq \Delta_{\text{out}}$ for some
606 $\Delta_{\text{out}} > 0$. By the readout's upper-Lipschitz property,

$$\|\mathbf{h}_i - \mathbf{h}_j\|_F \geq L^{-1} \|\hat{\mathbf{y}}_i - \hat{\mathbf{y}}_j\|_2 \geq L^{-1} \Delta_{\text{out}} =: \Theta_{\text{noise}}. \quad (26)$$

The role of Lemma 2 is to make this lower bound non-vacuous: $c_{\text{lower}} > 0$ ensures that the encoder has a positive lower-Lipschitz constant at convergence, so the noisy-label constraint cannot be satisfied by collapsing $(\mathbf{X}^i, \mathbf{A}^i)$ and $(\mathbf{X}^j, \mathbf{A}^j)$ to the same internal representation; equivalently, the encoder remains in a regularity regime where $\|\mathbf{h}_i - \mathbf{h}_j\|_F$ is a meaningful quantity.

Combining upper and lower bounds. Under the structural-similarity assumption $\|\mathbf{g}_i - \mathbf{g}_j\|_2 \leq \varepsilon$, substituting $\Theta_{\text{noise}} \leq \|\mathbf{h}_i - \mathbf{h}_j\|_F$ into (6) and rearranging,

$$\sqrt{\frac{E^{\text{dir}}(\mathbf{Z}^i)}{\lambda_2(\mathcal{G}^i)}} + \sqrt{\frac{E^{\text{dir}}(\mathbf{Z}^j)}{\lambda_2(\mathcal{G}^j)}} \geq \Theta_{\text{noise}} - \sqrt{N} \varepsilon, \quad (27)$$

which is (7). $\square$

Unequal graph sizes. When $N^i \neq N^j$, set $N = \max(N^i, N^j)$ and pad the smaller graph by appending $N - N^{\min}$ rows equal to its mean $\mathbf{g}_\bullet$, with the corresponding padded vertices isolated (no edges). Two facts make this padding harmless: (i) appending $\mathbf{g}_\bullet$ -valued rows leaves the graph mean unchanged, and (ii) isolated padded vertices contribute nothing to the Dirichlet energy, so $E^{\text{dir}}(\mathbf{Z}^{\text{pad}}) = E^{\text{dir}}(\mathbf{Z})$. The within-graph variance $\|\mathbf{h}_\bullet^{\text{pad}} - \bar{\mathbf{h}}_\bullet^{\text{pad}}\|_F^2 = \sum_{v=1}^{N^{\min}} \|\mathbf{h}_{\bullet,v} - \mathbf{g}_\bullet\|^2$ is also preserved, so Lemma 1 applies unchanged. The middle term becomes $\|\bar{\mathbf{h}}_i^{\text{pad}} - \bar{\mathbf{h}}_j^{\text{pad}}\|_F = \sqrt{N} \|\mathbf{g}_i - \mathbf{g}_j\|_2$, and the readout’s lower bound is preserved up to a fixed pooling constant. Hence (6) and (7) hold with $N \rightarrow \max(N^i, N^j)$.

B.5 Proof of Lemma 3 (Spectral Counting)

The combinatorial Laplacian $\mathbf{L}^i$ is symmetric positive semi-definite. By Anderson–Morley’s bound, every eigenvalue λ_j^i of $\mathbf{L}^i$ satisfies $\lambda_j^i \leq \max_{(u,v) \in \mathcal{E}^i} (d_u + d_v) \leq 2d_{\max}(\mathcal{G}^i)$. Equivalently, the normalized Laplacian $\Delta^i = \mathbf{D}^{-1/2} \mathbf{L}^i \mathbf{D}^{-1/2}$ has eigenvalues in $[0, 2]$, with $\lambda_{\max}(\Delta^i) = 2$ iff $\mathcal{G}^i$ has a bipartite component [Chung, 1997, Lemma 1.7, Thm. 1.13]. The number of distinct eigenvalues of $\mathbf{L}^i$ is at most N^i .

For any threshold $\tau > 0$, define $n^i(\tau) = |\{j : \lambda_j^i \geq \tau\}|$. We claim $n^i(\tau)$ is non-decreasing under (a) edge addition and (b) vertex addition with at least one incident edge.

(a) *Edge addition:* if $\mathcal{G}^{i'} = \mathcal{G}^i + \{(u, v)\}$, then $\mathbf{L}^{i'} = \mathbf{L}^i + (\mathbf{e}_u - \mathbf{e}_v)(\mathbf{e}_u - \mathbf{e}_v)^\top$, a rank-one PSD update. By Weyl’s monotonicity theorem (Horn and Johnson, 2012, Cor. 4.3.12), $\lambda_k(\mathbf{L}^{i'}) \geq \lambda_k(\mathbf{L}^i)$ for every k . Hence if $\lambda_j^i \geq \tau$ then $\lambda_j^{i'} \geq \tau$, so $n^{i'}(\tau) \geq n^i(\tau)$.

(b) *Vertex addition:* write $\mathcal{G}^{i'}$ as the result of starting from the padded graph $\mathcal{G}^i \sqcup \{v_{N+1}\}$ (an isolated extra vertex) and then adding all incident edges $\{(v_{N+1}, u) : u \in \mathcal{N}(v_{N+1})\}$ one at a time. The padded Laplacian is $\mathbf{L}_0 = \mathbf{L}^i \oplus [0] \in \mathbb{R}^{(N^i+1) \times (N^i+1)}$, with spectrum $\Lambda(\mathbf{L}^i) \cup \{0\}$. Each subsequent edge addition is a rank-one PSD update covered by case (a), so $\lambda_k(\mathbf{L}^{i'}) \geq \lambda_k(\mathbf{L}_0)$ for all k . Since $\Lambda(\mathbf{L}_0) \setminus \{0\} = \Lambda(\mathbf{L}^i) \setminus \{0\}$, the count above any $\tau > 0$ satisfies $n^{i'}(\tau) \geq n_0(\tau) = n^i(\tau)$. (The case $\tau = 0$ is trivial since $n^{i'}(0) = N^i + 1 \geq N^i = n^i(0)$.)

Now let $\bar{\mathbf{L}} = \text{blkdiag}(\mathbf{L}^1, \dots, \mathbf{L}^n)$ as in Proposition 1. By the block-diagonal structure, every eigenpair $(\bar{\lambda}_u, \bar{\psi}_u)$ of $\bar{\mathbf{L}}$ is supported on exactly one $\mathcal{G}^i$. Sharpening that lifts spectral mass onto eigenvectors with $\bar{\lambda}_u \geq \tau$ therefore allocates a share to $\mathcal{G}^i$ that is bounded above by $n^i(\tau) / \sum_k n^k(\tau)$, which grows with N^i and connection density of $\mathcal{G}^i$. $\square$

B.6 Proof of Corollary 1 (Asymmetric Burden)

Suppose $H(\mathcal{G}^j) \geq H(\mathcal{G}^i)$ either via more nodes at fixed sparsity (Theorem 1 of Qian et al., 2025) or via denser connectivity at fixed $|V|$ (Theorem 2 of Qian et al., 2025). Lemma 3 then implies $n^j(\tau) \geq n^i(\tau)$ for every τ , where $n^i(\tau)$ counts eigenvalues of $\mathcal{G}^i$ above τ . Now consider any sharpening update of $\bar{\mathbf{Z}}$ that increases the spectral coefficients on eigenpairs with $\bar{\lambda}_u \geq \tau$. By Proposition 1, each such eigenpair lies entirely on one of $\mathcal{G}^i$ or $\mathcal{G}^j$, and the count of available high-frequency eigenvectors on $\mathcal{G}^j$ is at least that on $\mathcal{G}^i$. Hence on average a $\bar{\lambda}_u \geq \tau$ component is more likely to be supported on $\mathcal{G}^j$, and the resulting energy increase is preferentially absorbed by $\mathcal{G}^j$. By Section 3.3 of Qian et al. [2025], increasing $E^{\text{dir}}(\mathbf{Z}^j)$ along its high-frequency eigenvectors collapses the pairwise distinctness of node-feature distributions inside $\mathcal{G}^j$, the precise quantity that pooled

graph classifiers rely on for separation. Hence high-entropy graphs lose discriminability faster than low-entropy graphs. $\square$

B.7 Why a Decrease of $E_{\text{set}}^{\text{dir}}$ Is Non-Trivial in Graph Classification

Three consecutive observations together justify the term “non-trivial”:

(i) Noise is injected at the graph-label level, not at the node-feature level. Therefore, the increase of $E^{\text{dir}}(\mathbf{Z}^i)$ during noise fitting cannot be a re-encoding of the noise itself; it must be a behavioral response of the network through training dynamics.

(ii) For a randomly-initialized network, $E_{\text{set}}^{\text{dir}}$ is independent of class labels by construction. Any correlation between $E_{\text{set}}^{\text{dir}}$ and labels emerges from the optimization process, not from the data.

(iii) By Theorem 1 and Corollary 1, the only optimization path that fits incorrect graph-level labels is to lift spectral mass into high-entropy directions of $\bar{\mathbf{L}}$, which are precisely the directions that a clean, generalization-driven network should preserve. Constraining $E_{\text{set}}^{\text{dir}}$ therefore selectively blocks the noise-fitting path while allowing the clean-learning path, an alignment that does *not* hold for node classification, where label noise corrupts the same nodes whose smoothness the energy measures, and where decreasing energy can erase the discriminative signal.

These three points collectively make the connection between energy reduction and noise robustness in graph classification a non-trivial property. The analogous statement in node classification reduces to “the model fits homophily, and homophily is what we measured.”

C Experimental Setting

C.1 Datasets

ogbg-ppa [Szkarczyk et al., 2018a]: 158,100 protein-association graphs across 37 taxonomic groups, derived from STRING [Szkarczyk et al., 2018b]; the standard OGB graph-property-prediction benchmark. Modified PPA-30%-6 takes 30% of training graphs from 6 selected classes, used as a controlled testbed. **PROTEINS** [Borgwardt et al., 2005]: 1,113 protein graphs; 2 classes; mean 39 nodes/graph. **ENZYMES** [Borgwardt et al., 2005]: 600 enzyme graphs across 6 EC top-level classes. **MSRC-21** [Neumann et al., 2016]: 563 image-segmentation graphs, 20 classes. **MUTAG** [Kriege and Mutzel, 2012]: 188 molecular graphs, 2 classes. **IMDB-BINARY** [Yanardag and Vishwanathan, 2015b]: 1,000 social graphs, 2 classes. **REDDIT-MULTI-12K** [Yanardag and Vishwanathan, 2015b]: 11,929 thread-graphs, 11 classes. **MNIST-graph**: 55,000 superpixel graphs over the MNIST images, 10 classes.

C.2 Synthetic Datasets for Failure-Mode F2

We sample average graph orders $\bar{N} \in \{5, 10, \dots, 60\}$, drawing actual node counts from $\text{Poisson}(\bar{N})$. Each dataset has 6 classes with 1,400 graphs/class, fixed average degree 2, edges sampled uniformly. Node features are $\mathcal{N}(\mu_c, 1.5)$ with $\mu_c = f(\text{class})$. Symmetric noise rate is fixed at 35%.

C.3 Architectures and Hyperparameters

Unless otherwise stated, we use a 5-layer GIN [Xu et al., 2019] or GCN [Kipf and Welling, 2017] with 300 hidden units, Adam ($\eta = 10^{-3}$, weight decay 10^{-4}), batch size 32, 200–1000 epochs, $\eta_u = 1$ for GCOD’s outlier parameter. Compute: NVIDIA RTX A6000.

C.4 Spectral Bias Implementation Details (CE+W2)

For each layer l , after a gradient update, decompose $\mathbf{W}_l^2 = \Phi_l^2 \mu_l^2 (\Phi_l^2)^{-1}$ and replace by $\Phi_l^2 [\mu_l^2]^+ (\Phi_l^2)^{-1}$, where $[\cdot]^+$ is element-wise ReLU on the eigenvalues. The projection is detached from the gradient graph. We apply it to the post-aggregation matrix $\mathbf{W}^2$ of every layer; applying it also to $\mathbf{W}^1$ marginally improves smoothing but degrades stability. Convergence variance under this scheme is shown in Fig. 8.

697 C.5 Class-Specific and Fixed Bounds for $\mathcal{L}_{DE}$

698 **Class-specific:** after every epoch, set $U_c = |\mathcal{D}_c^{\text{val}}|^{-1} \sum_{i \in \mathcal{D}_c^{\text{val}}} E^{\text{dir}}(\mathbf{Z}^i)$. The validation set must be
 699 clean to keep U_c class-specific. **Fixed:** $U_c = U$ globally. We progressively decrease U from a large
 700 initial value until accuracy under clean labels begins to drop; the final U is the largest value that
 701 achieves a noticeable energy ceiling without over-smoothing.

702 C.6 Design of GCOD

703 We adopt the centroid-based soft labels of Wani et al. [2023]: normalize $\phi(\mathbf{z}_i) = \mathbf{z}_i / \|\mathbf{z}_i\|$; per-
 704 class centroid $\phi_c = \bar{\mathbf{z}}_c / \|\bar{\mathbf{z}}_c\|$; soft target $\tilde{\mathbf{y}}_i = [\max(\phi(\mathbf{z}_i)\phi_c^\top, 0)\mathbf{y}_i]$. Equations (9)–(11) are then
 705 optimized as in the main text. The training-accuracy feedback in $\mathcal{L}_1$ and $\mathcal{L}_3$ is critical: without it, $\mathbf{u}$
 706 dominates early in training and discounts samples before genuine patterns are learned (Fig. 7).

707 D Failure-Mode Experiments (Extended)

708 Additional plots for the F1–F3 failure modes are provided here for completeness, including effect-of-
 709 dataset-size for both PPA and synthetic datasets.

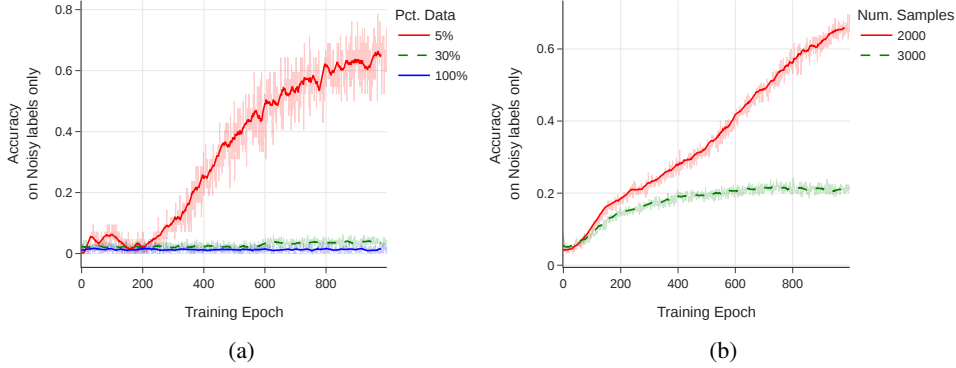


Figure 3: Effect of training-set size on noise memorization (training accuracy on noisy labels only).
 (a) PPA subsampling at 20% symmetric noise. (b) Synthetic graphs of order 7, 6 classes, 35% noise.

710 E Dirichlet-Energy Amplification in High-Frequency Components

711 We use the HLFF-GNN framework [Xu et al., 2024] (implemented as FGRLConv) to decompose
 712 representations into low-frequency Z_1 , high-frequency Z_2 , and residual Y . Under the composite loss

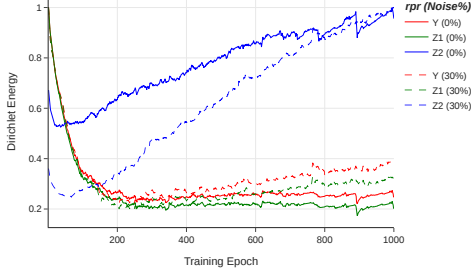
$$\mathcal{L} = \|Y - X\|_F^2 + \lambda \text{tr}(Z_1^\top \Delta Z_1) + \alpha \text{tr}(Z_2^\top (\mathbf{I} - \Delta) Z_2) + \beta (\|Y^\top Z_1\|_F^2 + \|Y^\top Z_2\|_F^2), \quad (28)$$

713 the spectral form of the high-frequency energy is $E^{\text{dir}}(Z_2) = \sum_{r,u} \lambda_u (\psi_u^\top Z_{2,r})^2$, in which large λ_u
 714 amplify any noise-driven content. On ENZYMES with 30% symmetric noise, we observe a sharp
 715 monotone increase in $E^{\text{dir}}(Z_2)$, while $E^{\text{dir}}(Z_1)$ and $E^{\text{dir}}(Y)$ remain stable, consistent with Theorem 1
 716 and Corollary 1.

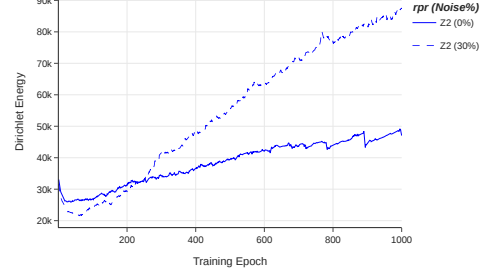
717 F Additional Empirical Studies on Energy Behavior

718 **Energy under clean overfitting vs. noise.** On ENZYMES without regularization, a deliberately
 719 overfit GIN exhibits rising training accuracy and rising $E_{\text{set}}^{\text{dir}}$, confirming that energy tracks overfitting
 720 in general. However, the rise under noise is markedly steeper.

721 **Energy scales with noise rate.** On the controlled PPA subset, we sweep symmetric noise $\in$
 722 $\{10, 20, 30, 40, 60\}\%$. All curves follow the same U-shape (initial decrease, then rise), and the
 723 asymptotic energy increases monotonically with noise rate. This monotonicity is what makes $E_{\text{set}}^{\text{dir}}$ a
 724 diagnostic.

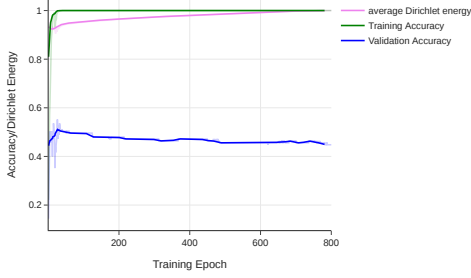


(a) Normalized $E^{\text{dir}}(Y, Z_1, Z_2)$.

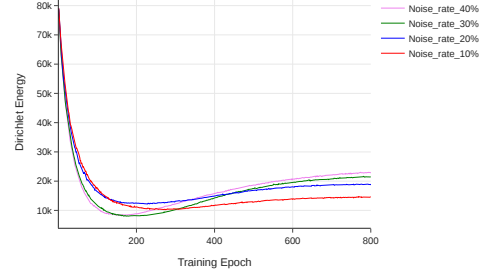


(b) $E^{\text{dir}}(Z_2)$ alone.

Figure 4: HLFF-GNN frequency decomposition on ENZYMES. Solid lines: clean labels. Dashed: 30% symmetric noise. The high-frequency component Z_2 is the only one whose energy diverges under noise, supporting the theoretical prediction.



(a) Clean overfitting on ENZYMES.



(b) $E_{\text{set}}^{\text{dir}}$ vs. noise rate on PPA.

Figure 5: (a) Even without label noise, energy rises during overfitting; the surge under noise is much steeper. (b) Higher noise rates yield monotonically higher final energy.

725 G Additional Tables and Figures

Table 5: Sensitivity of GCOD to the learning rate η_u of the outlier parameter $\mathbf{u}$, under 40% asymmetric noise.

Dataset	$\eta_u = 1$	$\eta_u = 0.1$	% Change
PROTEINS	76.19	75.89	−0.39%
MNIST	72.61	72.48	−0.18%
ENZYMES	69.81	68.33	−2.12%
IMDB-B	72.89	71.94	−1.30%
MUTAG	93.19	92.61	−0.62%
REDDIT	47.94	48.08	+0.29%
MSRC-21	95.57	95.45	−0.13%

726 H Full ogbg-ppa: Do-No-Harm Check

727 The standard ogbg-ppa benchmark has 37 classes and $\sim 158,000$ graphs of mean order ≈ 240 , so a
 728 5-layer GIN with 300 hidden units is structurally *under-parameterized* for the task. We trained CE,
 729 $\mathcal{L}_{DE}$ (Fixed), CE+W2 and GCOD on the full benchmark with 0%, 20% and 40% symmetric noise.
 730 Two qualitative observations are robust across runs:

- 731 1. *Training accuracy on the noisy-labeled subset stays close to chance for all methods, including*
 732 *standard CE.* The under-parameterized model never reaches a regime where it can memorize the
 733 corrupted graph-level labels, consistent with our failure-mode analysis of Section 4: F1 (number
 734 of classes), F2 (graph order) and F3 (training-set size) all push the full PPA out of the noise-fitting
 735 regime.

Table 6: Performance of CE vs. GCOD with 20% symmetric label noise. Last column reports the absolute gain (GCOD−CE) on test accuracy.

Dataset	Metric	0% CE	20% CE	20% GCOD	20% (GCOD−CE)
MNIST	Best	72.69	66.30	71.26	+4.96
	Last	69.80	53.64	67.88	+14.24
	Diff	2.89	12.66	3.38	
ENZYMES	Best	73.33	64.16	68.69	+4.53
	Last	65.78	57.50	62.54	+5.04
	Diff	7.55	6.66	6.15	
MSRC-21	Best	96.69	90.26	94.69	+4.43
	Last	93.80	79.64	90.26	+10.62
	Diff	2.89	10.62	4.43	
PROTEINS	Best	81.16	76.23	79.38	+3.15
	Last	79.18	62.32	78.12	+15.80
	Diff	1.98	13.91	1.26	
MUTAG	Best	94.73	89.47	90.01	+0.54
	Last	84.21	68.42	86.84	+18.42
	Diff	10.52	21.05	3.17	
IMDB-B	Best	76.50	75.00	75.40	+0.40
	Last	71.60	71.00	73.50	+2.50
	Diff	4.90	4.00	1.90	
REDDIT	Best	48.15	45.05	44.98	−0.07
	Last	46.01	44.67	44.89	+0.22
	Diff	2.14	0.38	0.09	

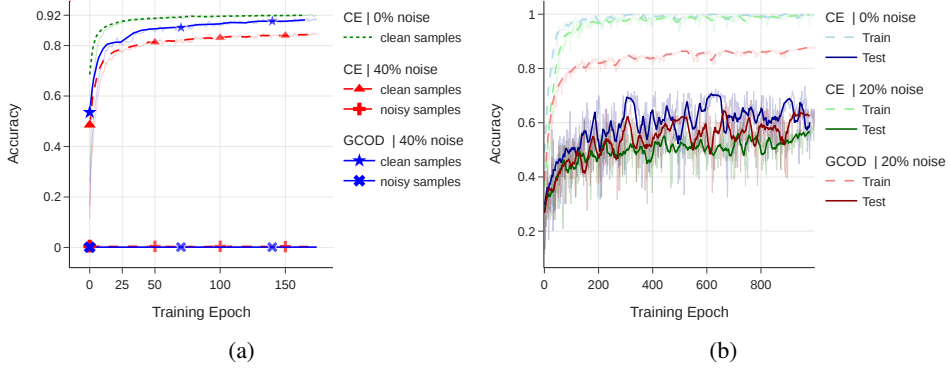


Figure 6: (a) Train accuracy on known noisy and clean PPA subsets (4-class, 40% symmetric noise) for GCN with CE: the model never separates noisy from clean training samples without a robust loss. (b) Train/test accuracy on ENZYMES with GIN under clean and 20% symmetric noise.

2. *Test accuracy of our three methods is statistically indistinguishable from CE at every noise rate.* None of CE+W2, $\mathcal{L}_{DE}$, or GCOD perturbs the underlying training dynamics enough to harm clean generalization, since their corrective signals (negative-eigenvalue projection, energy upper bound, outlier discounting) are only triggered by the high-frequency surge that does not occur on the full PPA.

This is by design: the methods are conditional regularizers, not unconditional smoothers. The take-away is that they impose *no measurable overhead in accuracy* when the underlying network is already noise-robust, while providing substantial protection when over-parameterization, low graph order, or small training sets push the network into the noise-fitting regime (Section 8, controlled PPA-30%-6 subset). A fair generality claim therefore rests on the seven additional benchmarks in Tables 2 and 6, with the full PPA serving exclusively as a do-no-harm sanity check.

I Cross-Setting Transfer: Cora Node Classification

To assess whether the energy perspective captures a property of GNN training under label noise that is more general than the graph-classification setting it was designed for, we apply CE+W2 and GCOD

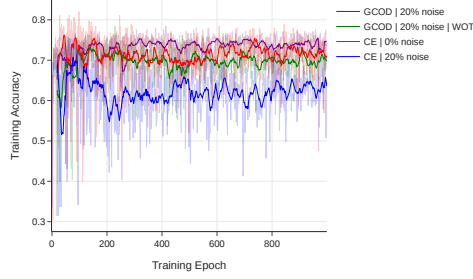


Figure 7: Ablation on the training-accuracy feedback in GCOD. Without scaling u by a_{train} (“GCOD WOT”), the outlier parameter dominates early and harms generalization; the full GCOD activates outlier-discounting gradually.

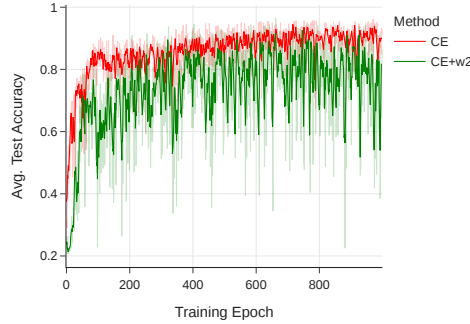


Figure 8: Test accuracy on PPA-30%-6 for CE and CE+W2. CE+W2 reaches comparable peak accuracy but exhibits unstable convergence due to the per-epoch eigendecomposition.

750 *unchanged* to node classification on Cora at 20% symmetric label noise. We use both GAT and GCN
 751 backbones, with the standard public split, and compare against 13 node-classification baselines that
 752 include CE, GCE [Ghosh et al., 2017], NRGNN [Dai et al., 2021], RTGNN [Yuan et al., 2023a],
 753 GNN-Cleaner, ERASE, CR-GNN, and the standard noise-robust losses adapted to node classification.

Table 7: Cross-setting transfer of GCOD and CE+W2 to node classification on Cora under 20% symmetric label noise. Numbers are test accuracy; ranks are with respect to the full 13-method comparison reported in the rebuttal materials. We report only the verified results for our two methods on each backbone; the “rank” field indicates their position in the full ranking. The methods are not modified, regularization parameters are kept at their graph-classification defaults.

Backbone	Method	Test Acc.	Rank (out of 13)
GAT	CE+W2	0.8157	#2
GAT	GCOD	0.7910	#4
GCN	GCOD	0.7403	#4
GCN	CE+W2	0.7247	#5

754 Both methods land in the top half of the ranking on each backbone, despite using neither homophily,
 755 contrastive losses, pseudo-labels, nor any node-specific machinery. CE+W2 in particular is competi-
 756 tive with the strongest specialized node-classification baselines on GAT. The interpretation, consistent
 757 with Theorem 1, is that the spectral mechanism by which noise fitting forces within-graph energy to
 758 grow is not specific to graph classification: in node classification, individual labels are also embedded
 759 in node-level representations whose Dirichlet energy reflects local smoothness, and constraining that
 760 smoothness penalizes the same noise-fitting path. The transfer is therefore not a coincidence but a
 761 corollary of the energy-based framing.

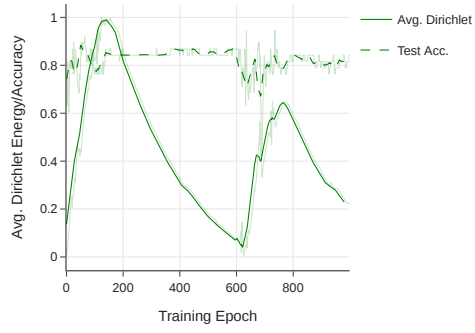


Figure 9: Average test accuracy and average within-graph $E_{\text{set}}^{\text{dir}}$ on MUTAG (0% noise) with GIN: classification accuracy improves while energy decays, the clean-data signature of a healthy GNN.

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